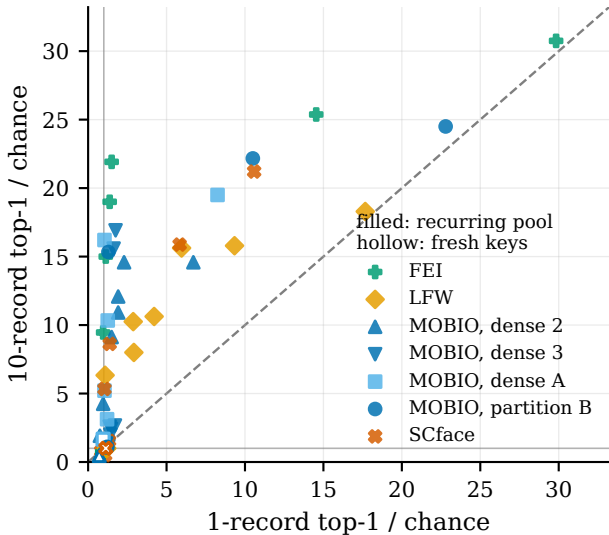
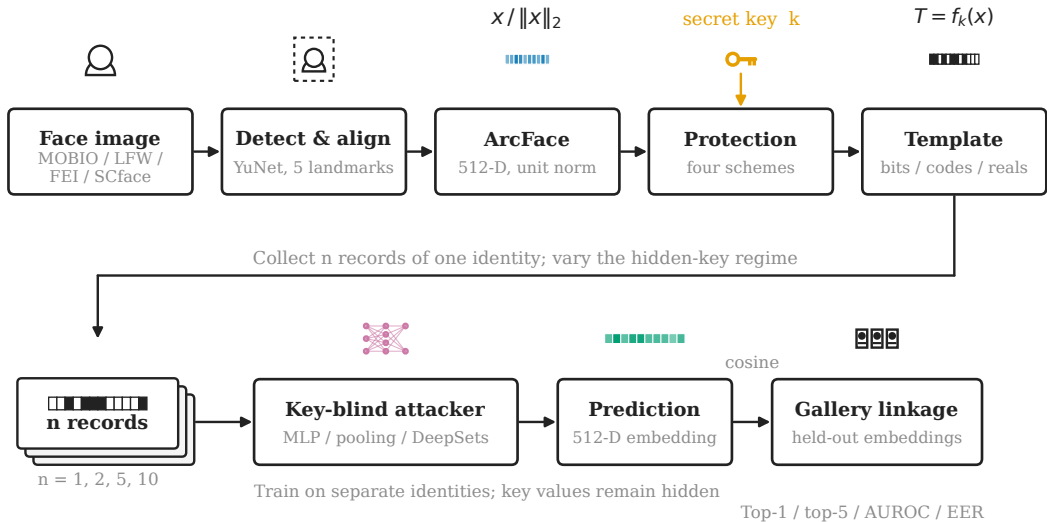


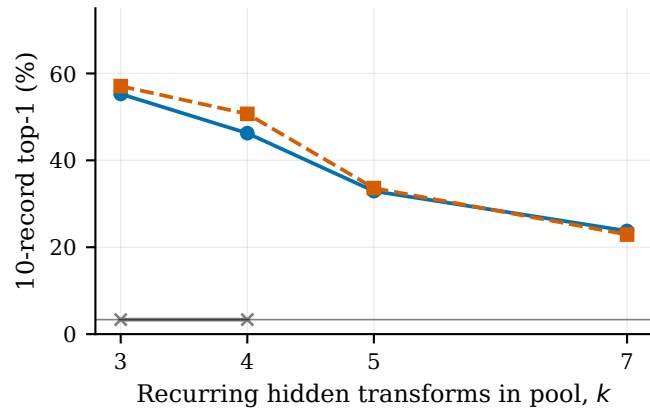
BioHash record-count gains





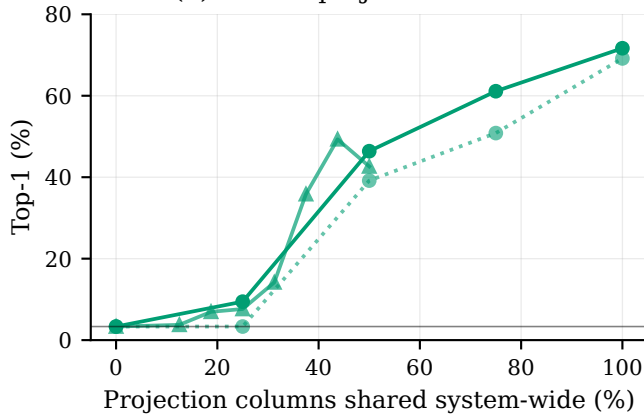
MOBIO mechanism controls (30 gallery identities; chance 3.33%)

(a) Slot label vs. shuffled records



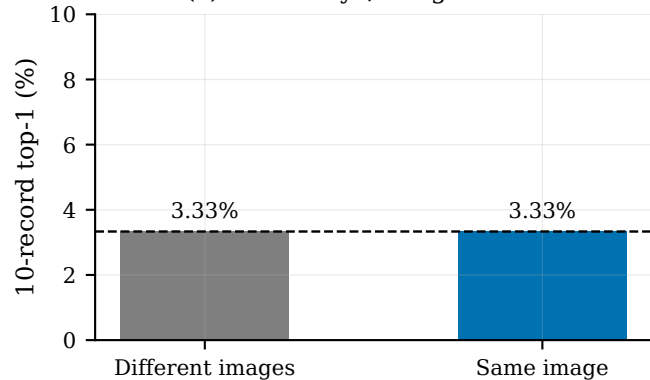
- slot hidden (DeepSets)
- slot label given to attacker
- records 2-10 from other identities

(b) Shared projection columns

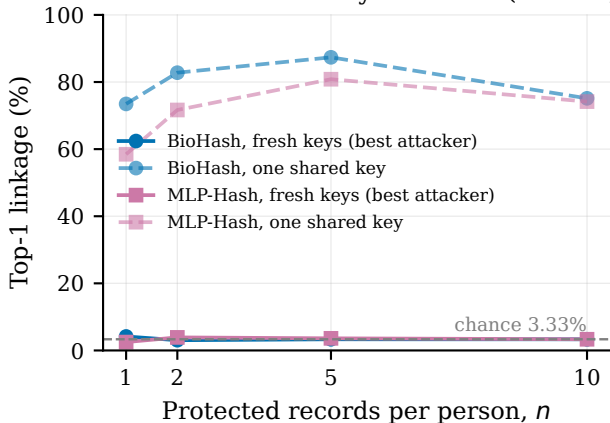


- 10 records, coarse sweep
- 10 records, fine sweep (new split)
- 1 record, coarse

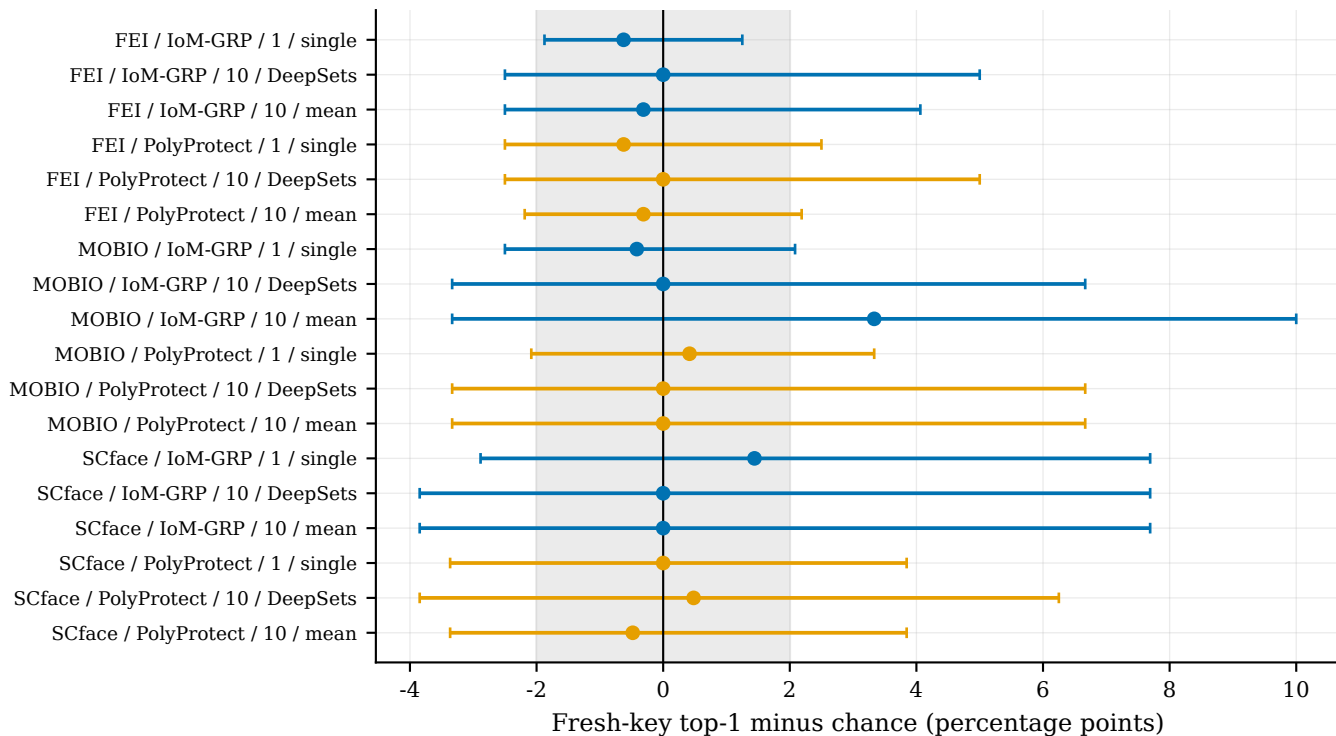
(c) Fresh keys, image control



Fresh and shared keys: MOBIO (N = 30)

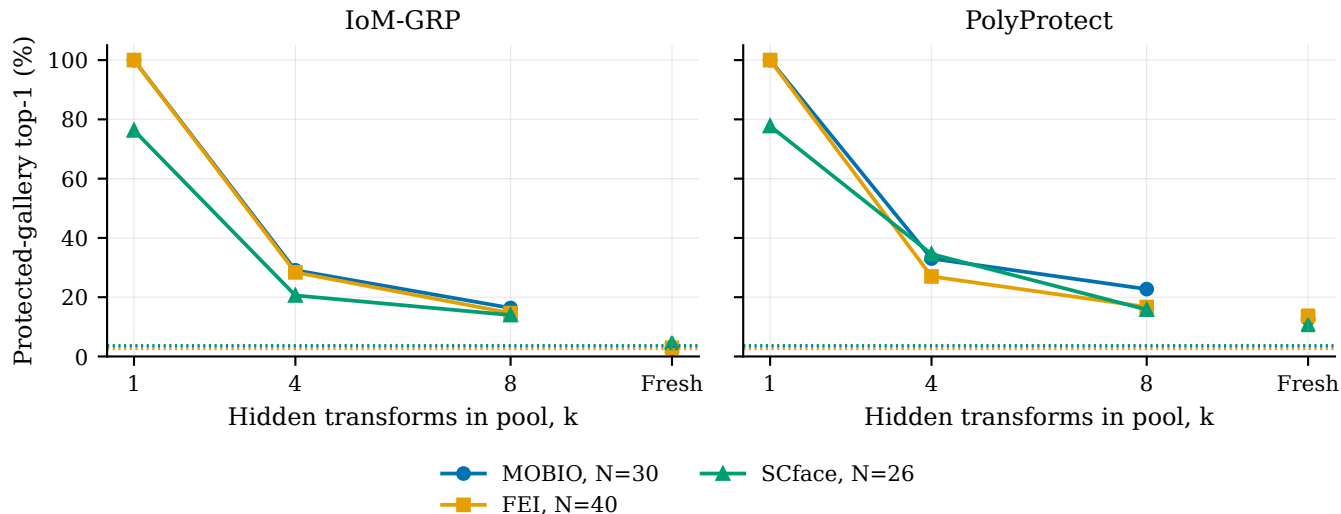


Fresh-key 90% intervals: exploratory pilot sensitivity

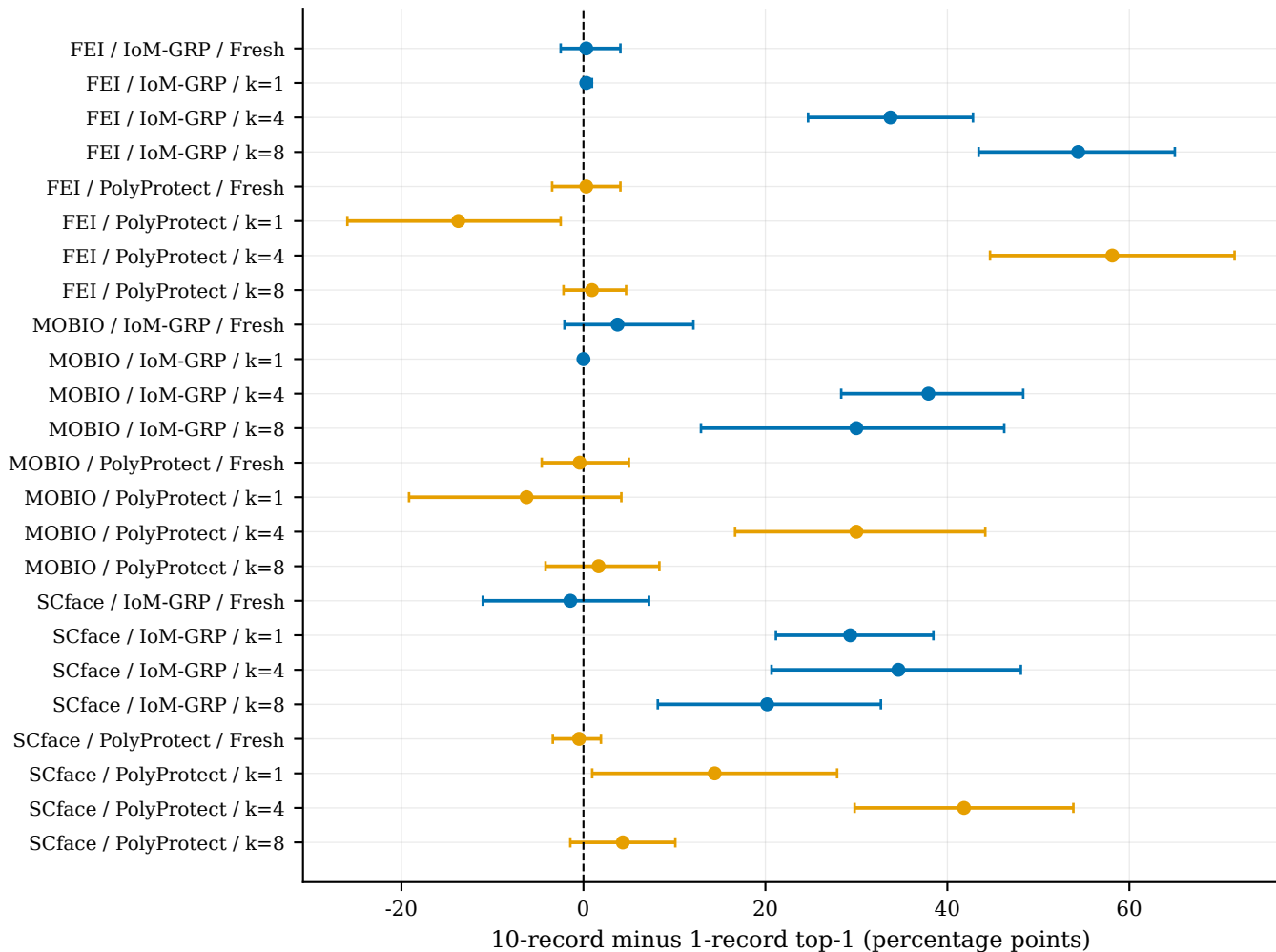


Illustrative +/-2 pp band; not an approved equivalence margin

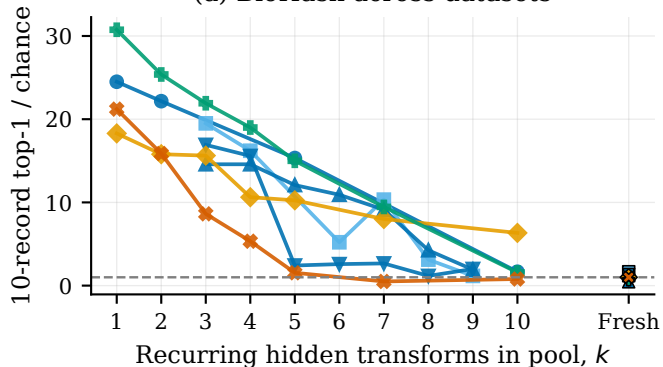
Native matching diagnostic, not learned embedding linkage



Paired mean-pool gains: one-seed pilots, 95% intervals

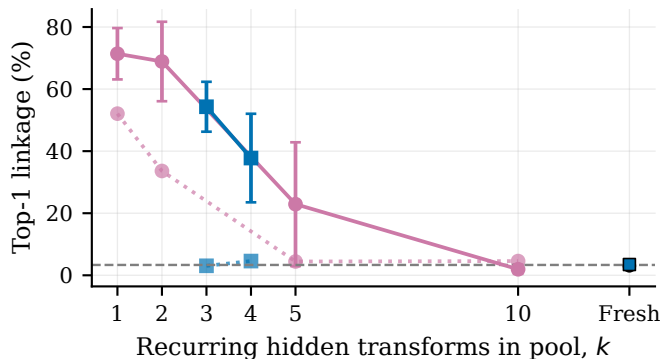


(a) BioHash across datasets



- MOBIO, partition B (n=30)
- MOBIO, dense A (n=30)
- ▲ MOBIO, dense 2 (n=30)
- ▼ MOBIO, dense 3 (n=30)
- ◆ LFW (n=25)
- ◆ FEI (n=40)
- ◆ SCface (n=26)

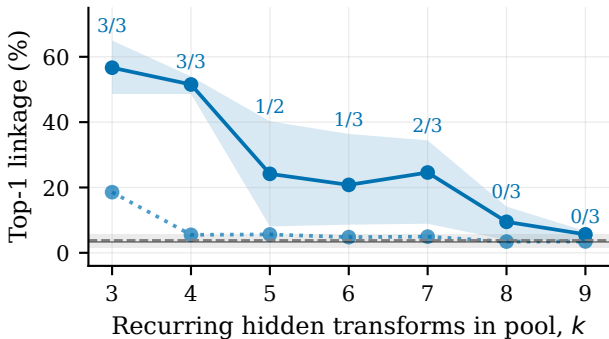
(b) MLP-Hash, MOBIO (N = 30)



Bars: seed SD, not confidence intervals

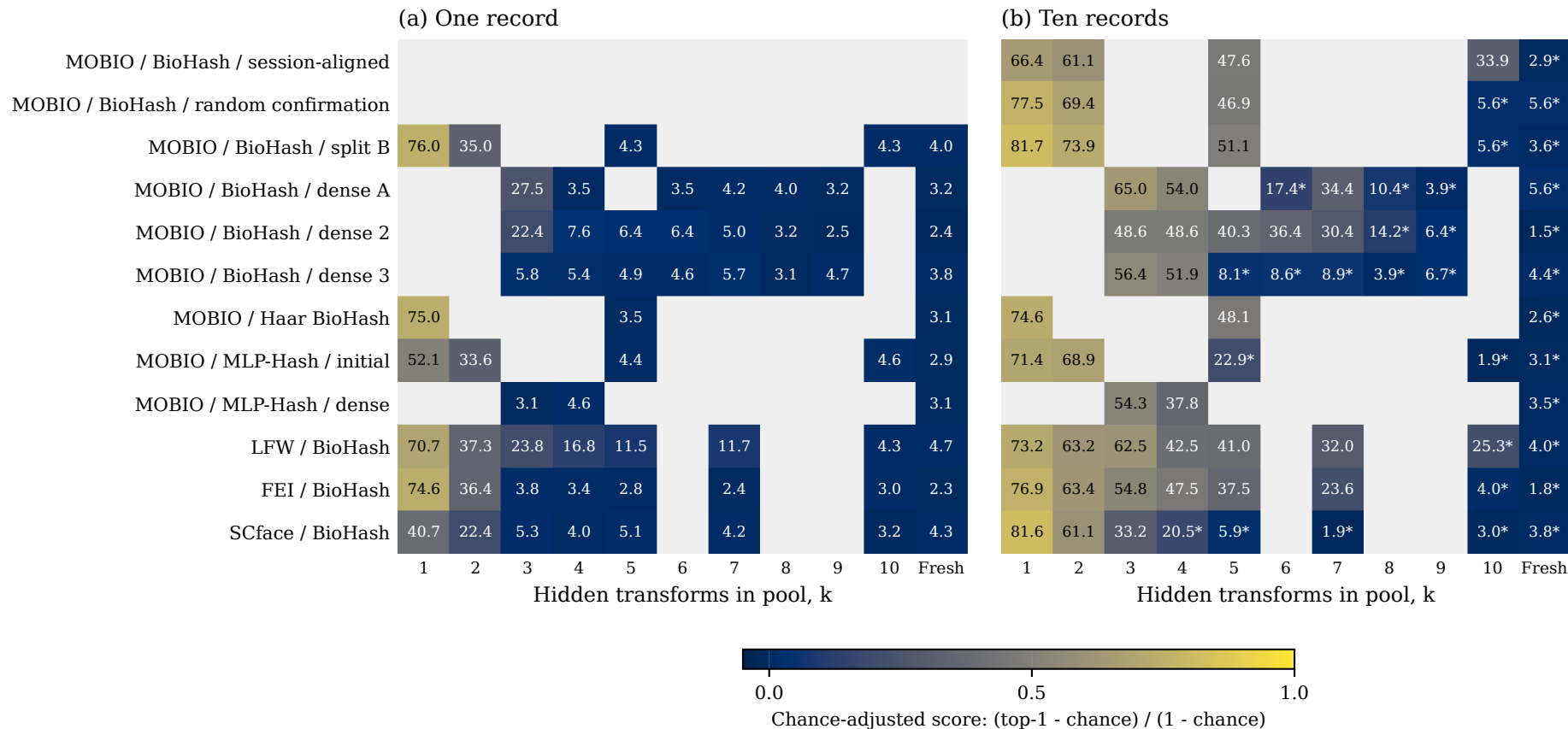
- Initial study, 1 record
- Dense study, 1 record
- Initial study, 10 records
- Dense study, 10 records

MOBIO boundary (N = 30)



- range over available partitions
- 10 records, pooled mean
- 1 record, pooled mean
- fresh keys (3 partitions)
- chance 3.33%

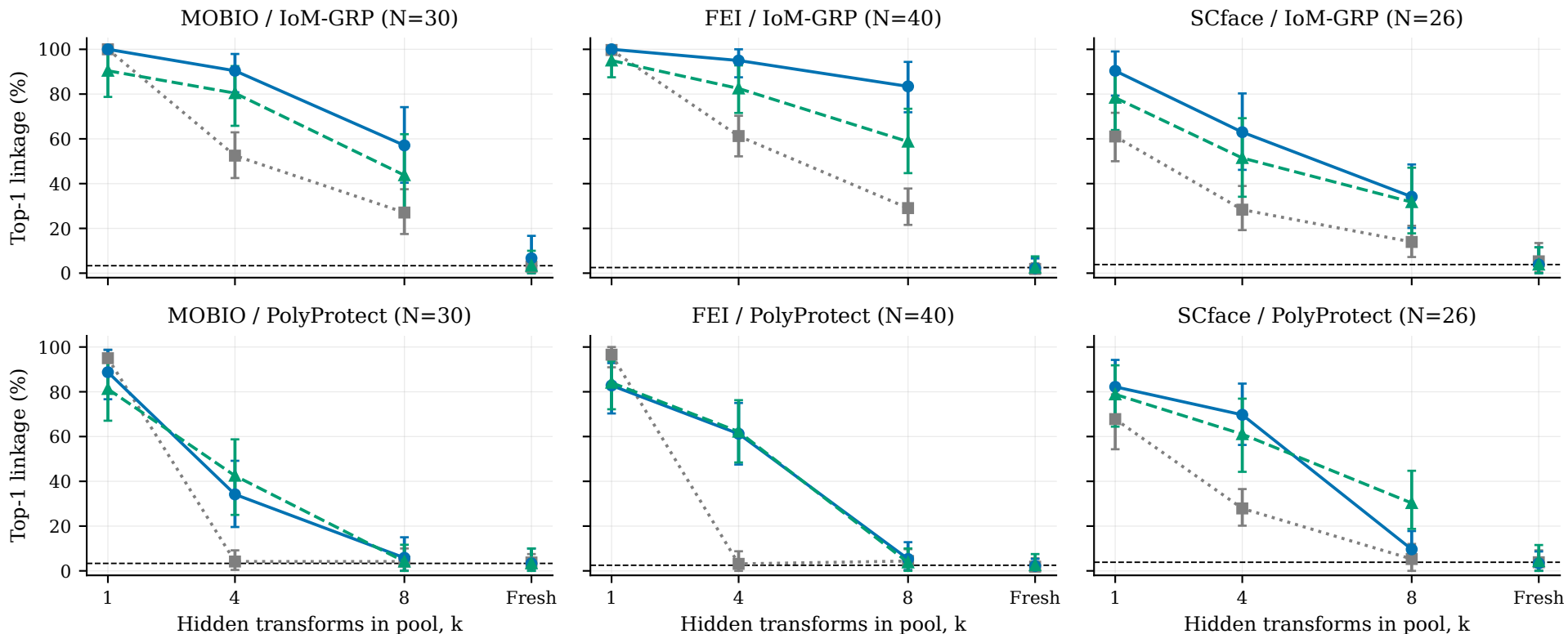
Earlier three-seed key-pool studies



Cells: top-1 (%), 3-seed means. Grey: unavailable. *: not all seed intervals exclude chance.
 Gallery sizes: MOBIO 30; LFW 25; FEI 40; SCface 26. Rows are separate studies, not matched replications.

New protection schemes: one-seed pilots

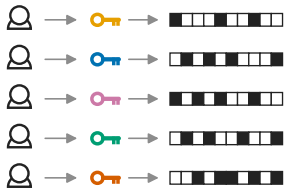
120-epoch cap; bars: identity-clustered 95% intervals; dashed baseline: chance



1 record 10 records, mean 10 records, DeepSets

Fresh keys

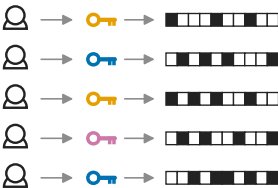
one key per source record



no transform reuse

Recurring key pool

k transforms; slot labels hidden

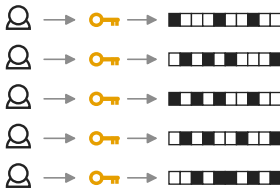


same colour = same transform

reuse across train and test

Single shared key

k = 1; key value still hidden



shared-transform baseline

Learned attacks inconclusive

PolyProtect outside the theorem

Multi-record gains observed

boundary depends on dataset and split

Strong linkage observed

hidden does not mean unlearnable

Protocol

Disjoint train / validation / test identities; one held-out gallery image per identity
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals